

Introduction and learning objectives

This tutorial builds on Lecture 5's analysis of validation-based model selection and Lecture 6's finite-class PAC bounds. The goal is to understand how performance on a sampled dataset supports guarantees on new examples, and how those guarantees change between the realizable and agnostic settings.

By the end of the tutorial, you should be able to:

- Interpret Lecture 6's PAC guarantees: distinguish training error, true error and excess risk, and explain ε and δ .
- Apply the lectures' Hoeffding and union-bound argument to every hypothesis, including one selected using the data.
- Explain Lecture 6's realizable-case argument and compare its $1/\varepsilon$ sample-size dependence with the general $1/\varepsilon^2$ bound.
- Use the lecture bounds to calculate sufficient sample sizes and excess-risk guarantees, explaining the role of repeated inputs.

1 PAC Bound: Realizable Case

Suppose we have binary classification on 3-bit inputs $x \in \{0, 1\}^3$, with labels $t \in \{0, 1\}$. The hypothesis class $\mathcal{H}$ contains all possible labelings of these 8 inputs and is fixed before observing the data. The dataset $\mathcal{D} = \{(x_i, t_i)\}_{i=1}^N$ comprises N i.i.d. samples from a fixed distribution $p(x, t)$. Using zero-one loss, write

$$L_{\mathcal{D}}(h) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{h(x_i) \neq t_i\}, \quad L_p(h) = \Pr_{(x,t) \sim p}(h(x) \neq t).$$

Assume realizability: there exists $h^* \in \mathcal{H}$ with $L_p(h^*) = 0$. Our learner returns an empirical risk minimizer $\hat{h} \in \arg \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)$, so $L_{\mathcal{D}}(\hat{h}) = 0$ with probability one.

Problem 1.1. How large is $\mathcal{H}$? (Recall from lecture that each of the 8 inputs can be labeled 0 or 1.)

Problem 1.2. For $0 < \varepsilon, \delta < 1$, a sufficient sample size for the finite hypothesis class PAC bound in the realizable case is:

$$N \geq \frac{1}{\varepsilon} \ln \frac{|\mathcal{H}|}{\delta}.$$

Explain the resulting guarantee on $L_p(\hat{h})$, including what the probability is over. How does this relate to the gap between training and true generalization error when $L_{\mathcal{D}}(\hat{h}) = 0$?

Problem 1.3. Plug in $\varepsilon = 0.1$ and $\delta = 0.05$. What is the smallest integer N satisfying this sufficient bound? Compare this to the number of distinct inputs (8). Why can this bound exceed the number of possible inputs?

2 Agnostic (Non-Realizable) Case

Now drop the realizability assumption, retaining the same input space, hypothesis class, i.i.d. sampling and zero-one loss. Labels may be noisy: for an input with positive probability, both labels may occur. Thus even a class containing all deterministic labelings need not contain a hypothesis with zero true error. Let

$$h^* \in \arg \min_{h \in \mathcal{H}} L_p(h)$$

be the best-in-class hypothesis. Our learning algorithm outputs:

$$\hat{h} \in \arg \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)$$

In the agnostic setting, zero true error is not assumed to be achievable. Instead, we measure the *excess risk* of $\hat{h}$:

$$L_p(\hat{h}) - L_p(h^*),$$

i.e. how much worse our learner is compared to the best-in-class h^* . Here ε denotes the uniform deviation between empirical and true error in Problem 2.1. The PAC bound shows that with high probability the excess risk is at most 2ε . Thus, the guarantee is not about achieving small absolute error, but about controlling the *gap* to the best possible hypothesis in $\mathcal{H}$.

Problem 2.1. For $0 < \delta < 1$, show that with probability at least $1 - \delta$ over the draw of $\mathcal{D}$:

$$\forall h \in \mathcal{H} : |L_{\mathcal{D}}(h) - L_p(h)| \leq \varepsilon,$$

where

$$\varepsilon = \sqrt{\frac{1}{2N} \ln \frac{2|\mathcal{H}|}{\delta}}.$$

(Hint: Hoeffding's inequality + union bound.)

Problem 2.2. Use this to prove:

$$L_p(\hat{h}) \leq L_p(h^*) + 2\varepsilon.$$

Problem 2.3. Compute ε for $N = 200$, $\delta = 0.05$, $|\mathcal{H}| = 256$, and state the resulting excess-risk bound.